Extremally projective normed spaces

N. T. Nemesh

Abstract: A normed space is called extremally projective if it is C-topologically projective in the category of normed spaces for every $C > 1$. Helemskii proved that the spaces $\ell_1^0(\Lambda)$ of finitely supported functions on a set Λ with the ℓ_1-norm are extremally projective and asked whether every extremally projective normed space is isometrically isomorphic to some $\ell_1^0(\Lambda)$. We answer this question in the negative. Then we show that the completion of an extremally projective normed space is isometrically isomorphic to $\ell_1(\Gamma)$ for some set Γ, and that extremally projective normed spaces contain no infinite dimensional complete subspaces; in particular, no infinite dimensional Banach space is extremally projective in the category of normed spaces. For normed spaces of countable algebraic dimension these necessary conditions turn out to be sufficient: such a space P is extremally projective if and only if its completion is isometrically isomorphic to $\ell_1(\Gamma)$ for a countable set Γ, if and only if the Banach-Mazur distance between P and $\ell_1^0(\Gamma)$ equals 1. Thus, extremal projectivity is an almost isometric rather than an isometric property. In particular, every dense subspace of ℓ_1 of countable dimension is extremally projective, and a separable normed space is extremally projective if and only if it has countable algebraic dimension and its completion is isometrically isomorphic to ℓ_1 or to ℓ_1^n.

Keywords: projective normed space, topological projectivity, extremal projectivity, Banach-Mazur distance, ℓ_1-space.

1 Introduction

Projectivity in categories of normed and Banach modules admits several non-equivalent versions. Besides the classical relative projectivity of Helemskii [1], which ignores the norms of lifting operators, there are so-called metric and topological projectivity [2, 4, 5], where the norm of a lifting is controlled by a constant. In the metric theory this constant is 1, and in the topological theory it is some fixed $C \geq 1$. The present paper is devoted to the limiting case between these two theories: a module is called extremally projective if it is C-topologically projective for every $C > 1$. This notion was introduced and studied by Helemskii under the name of extreme projectivity [3, 2]. We restrict ourselves to the simplest possible algebra, namely the zero algebra. Normed modules over the zero algebra are just normed spaces, so we deal with extremal projectivity in the category **Nor** of normed spaces and bounded linear operators.

For Banach spaces the corresponding questions are classical. By a theorem of Köthe [6] topologically projective Banach spaces are exactly the Banach spaces topologically isomorphic to $\ell_1(\Lambda)$ for some set Λ, and by a theorem of Grothendieck [7] a Banach space has the metric lifting property if and only if it is isometrically isomorphic to some $\ell_1(\Lambda)$. For normed spaces the answers are different, because the category **Nor** has its own free objects. These are the spaces $\ell_1^0(\Lambda)$ of finitely supported functions on a set Λ with the ℓ_1-norm. Grønbæk [9] extended Köthe's theorem to **Nor**: a normed space E is topologically projective if and only if it is topologically isomorphic to $\ell_1^0(\Lambda)$, where Λ is a set whose cardinality is the density character of E. Helemskii [2] proved the metric counterpart: a normed space is metrically projective if and only if it is isometrically isomorphic to some $\ell_1^0(\Lambda)$. For extremal projectivity the following facts were known. The spaces $\ell_1^0(\Lambda)$ are extremally projective, and a normed space is extremally projective if and only if it is a near-retract

of some $\ell_1^0(\Lambda)$ [3, propositions 2.2 and 2.3]. Every extremally projective normed space is, up to an isometric isomorphism, a dense subspace of $\ell_1(\Lambda)$ for some set Λ [3, proposition 2.3], see also [9, section 3], and every separable topologically projective normed space has at most countable algebraic dimension [3, proposition 2.4]. Both Helemskii [3, section 2] and Grønbæk [9, conjecture 3.1] asked whether, conversely, every extremally projective normed space is isometrically isomorphic to some $\ell_1^0(\Lambda)$; in a more detailed form, Helemskii asked which dense subspaces of $\ell_1(\Lambda)$ are extremally projective.

The main results of this paper are the following. First, the answer to the question of Helemskii and Grønbæk is negative: the subspace of ℓ_1 spanned by the finitely supported sequences and a single sequence with infinite support is extremally projective but not isometrically isomorphic to any $\ell_1^0(\Lambda)$. Second, for normed spaces of countable algebraic dimension extremal projectivity turns out to be an almost isometric property: such a space P is extremally projective if and only if its completion is isometrically isomorphic to $\ell_1(\Gamma)$ for a countable set Γ, if and only if for every $\varepsilon > 0$ there is a linear bijection $T : P \to \ell_1^0(\Gamma)$ with $\|T\|\|T^{-1}\| < 1 + \varepsilon$. In particular, every dense subspace of ℓ_1 of countable algebraic dimension is extremally projective. Combined with the result of Helemskii on separable topologically projective spaces, this gives a complete description of separable extremally projective normed spaces: these are exactly the normed spaces of countable algebraic dimension whose completions are isometrically isomorphic to ℓ_1 or to ℓ_1^n. Third, we extend the dimension restriction of Helemskii to non-separable spaces: every separable subspace of a topologically projective normed space has countable algebraic dimension, and therefore topologically projective normed spaces contain no infinite dimensional subspaces which are complete with respect to a norm dominating the original one. In particular, an infinite dimensional Banach space is never extremally projective in **Nor**, although it may well be extremally projective in the category **Ban** of Banach spaces. The last fact explains the main technical difficulty of the subject: when the lifting problems are posed for all normed spaces rather than Banach spaces, the standard tools of Banach space geometry, which rely on completeness, are no longer available, and we must work with the finite dimensional pieces of our spaces directly. The non-separable case remains open.

Let us fix the notation and recall the definitions. All normed spaces are considered over the complex field; the real case is identical. For a normed space E we denote by B_E its closed unit ball and by B_E° its open unit ball. By **Nor** we denote the category of normed spaces and bounded linear operators and by **Ban** its full subcategory of Banach spaces. For a set Λ we denote by $\ell_1(\Lambda)$ the Banach space of absolutely summable families and by $\ell_1^0(\Lambda)$ its dense subspace consisting of finitely supported families. For $\lambda \in \Lambda$ the symbol e_λ denotes the family which equals 1 at λ and 0 elsewhere; the family $(e_\lambda)_{\lambda\in\Lambda}$ is a Hamel basis of $\ell_1^0(\Lambda)$. For a normed space E and a set Λ there is a bijection between bounded families $(x_\lambda)_{\lambda\in\Lambda}$ in E and bounded linear operators $T : \ell_1^0(\Lambda) \to E$, given by $T(e_\lambda) = x_\lambda$, and $\|T\| = \sup_{\lambda\in\Lambda}\|x_\lambda\|$ [3, proposition 2.1], [9, proposition 1.10]. This is the sense in which the spaces $\ell_1^0(\Lambda)$ are free objects of **Nor**; note that no completeness of E is required here, in contrast with the case of $\ell_1(\Lambda)$.

Following [5], we say that a bounded linear operator $\xi : X \to Y$ between normed spaces is c-topologically surjective, where $c \geq 1$, if $c\,\xi(B_X^\circ) \supset B_Y^\circ$, and that ξ is coisometric if it is contractive and 1-topologically surjective, that is, if $\xi(B_X^\circ) = B_Y^\circ$. Equivalently, ξ is c-topologically surjective if for every $y \in Y$ and every $\delta > 0$ there exists $x \in X$ with $\xi(x) = y$ and $\|x\| < c\|y\| + \delta$. A linear bijection $T : E \to F$ is called a c-isomorphism if $\|T\|\|T^{-1}\| \leq c$. The Banach-Mazur distance between E and F is

$$d_{BM}(E, F) = \inf\{\|T\|\|T^{-1}\| : T : E \to F \text{ is a linear bijection}\},$$

with $d_{BM}(E,F) = +\infty$ if there are no bounded linear bijections with bounded inverses between E and F. A bounded linear operator $\zeta : E \to F$ is called a c-retraction if there exists a bounded linear operator $\eta : F \to E$ with $\zeta\eta = 1_F$ and $\|\zeta\|\|\eta\| \le c$. In this case F is called a c-retract of E.

Definition 1.1. *Let $C \ge 1$. A normed space P is called C-topologically projective if for every $c \ge 1$, every c-topologically surjective operator $\xi : X \to Y$ between normed spaces and every bounded linear operator $\phi : P \to Y$ there exists a bounded linear operator $\psi : P \to X$ such that $\xi\psi = \phi$ and $\|\psi\| \le cC\|\phi\|$. A normed space is called topologically projective if it is C-topologically projective for some $C \ge 1$, and extremally projective if it is C-topologically projective for every $C > 1$.*

The operator ψ in definition 1.1 is called a lifting of ϕ along ξ. Extremal projectivity is precisely the extreme projectivity of Helemskii, see proposition 2.4 below. We emphasize that the spaces X and Y in definition 1.1 are arbitrary normed spaces. If we restrict them to Banach spaces, we obtain the notions of topological and extremal projectivity in **Ban**, which we shall mention only for comparison. The two notions differ substantially, see section 3.

The paper is organized as follows. In section 2 we collect the basic properties of extremally projective normed spaces: the spaces $\ell_1^0(\Lambda)$ are extremally projective, extremal projectivity passes to $(1+\varepsilon)$-retracts, and a normed space is extremally projective if and only if it is a $(1+\varepsilon)$-retract of a free normed space for every $\varepsilon > 0$. In section 3 we discuss the dimension restrictions on topologically projective normed spaces. Section 4 contains the counterexample to the question of Helemskii and Grønbæk. Section 5 is devoted to the almost isometric characterization of extremally projective normed spaces of countable dimension and to the discussion of the non-separable case.

2 Basic properties of extremally projective normed spaces

We start with the free objects of **Nor**. The following proposition is [3, proposition 2.2] with explicit constants; we include the short proof for completeness.

Proposition 2.1. *Let Λ be a set. Then $\ell_1^0(\Lambda)$ is $(1+\varepsilon)$-topologically projective for every $\varepsilon > 0$. In particular, $\ell_1^0(\Lambda)$ is extremally projective.*

Proof. Let $\xi : X \to Y$ be a c-topologically surjective operator and $\phi : \ell_1^0(\Lambda) \to Y$ be a bounded linear operator. Fix $\varepsilon > 0$. For every $\lambda \in \Lambda$ put $y_\lambda = \phi(e_\lambda)$, so $\|y_\lambda\| \le \|\phi\|$. If $y_\lambda = 0$, put $x_\lambda = 0$. Otherwise $(1+\varepsilon)^{-1}\|y_\lambda\|^{-1}y_\lambda \in B_Y^\circ$, so there exists $x'_\lambda \in B_X^\circ$ with $c\,\xi(x'_\lambda) = (1+\varepsilon)^{-1}\|y_\lambda\|^{-1}y_\lambda$. Put $x_\lambda = c(1+\varepsilon)\|y_\lambda\|x'_\lambda$, then $\xi(x_\lambda) = y_\lambda$ and $\|x_\lambda\| < c(1+\varepsilon)\|y_\lambda\| \le c(1+\varepsilon)\|\phi\|$. Since the family $(x_\lambda)_{\lambda\in\Lambda}$ is bounded, there is a bounded linear operator $\psi : \ell_1^0(\Lambda) \to X$ with $\psi(e_\lambda) = x_\lambda$ for all $\lambda \in \Lambda$ and $\|\psi\| = \sup_\lambda\|x_\lambda\| \le c(1+\varepsilon)\|\phi\|$. Finally, $\xi\psi$ and ϕ coincide on the Hamel basis $(e_\lambda)_{\lambda\in\Lambda}$, so $\xi\psi = \phi$. □

The constant $1+\varepsilon$ in proposition 2.1 cannot be replaced by 1. In fact, with the convention of definition 1.1, 1-topological projectivity is an empty notion in **Nor**.

Proposition 2.2. *The only 1-topologically projective normed space is the zero space.*

Proof. Let X be the space of finitely supported sequences with the sup-norm and $f : X \to \mathbb{C}$ be the functional $f(x) = s^{-1}\sum_{n=1}^\infty 2^{-n}(1-2^{-n})x_n$, where $s = \sum_{n=1}^\infty 2^{-n}(1-2^{-n})$. Clearly, $\|f\| \le 1$. The operator f is coisometric: for $z \in B_\mathbb{C}^\circ$ choose N with $|z| < s^{-1}\sum_{n=1}^N 2^{-n}(1-2^{-n})$, then $z = f(x)$ for the sequence $x = t\,\mathrm{sgn}(z)\sum_{n=1}^N e_n$ with a suitable $t \in [0,1)$, and $\|x\| < 1$. On the

other hand, $|f(x)| = 1$ for $x \in B_X$ would force $x_n = \operatorname{sgn}(f(x))$ for all n, which is impossible for a finitely supported sequence. Thus, $f(B_X) = f(B_X^\circ) = B_{\mathbb{C}}^\circ$.

Now let P be a non-zero 1-topologically projective normed space. Take $p \in P$ with $\|p\| = 1$ and, by the Hahn-Banach theorem, a functional $g \in P^*$ with $\|g\| = g(p) = 1$. Lifting g along the coisometric operator f we get an operator $\psi : P \to X$ with $f\psi = g$ and $\|\psi\| \leq 1$. Then $x = \psi(p) \in B_X$ and $f(x) = g(p) = 1$, which contradicts the previous paragraph. □

Proposition 2.2 shows why in **Nor** it is natural to consider all constants $C > 1$ at once, instead of the single constant $C = 1$ of the metric theory.

Remark 2.3. *The proof of proposition 2.2 uses only the fact that the coisometric functional f does not attain its norm. Functionals of this kind exist on Banach spaces too, for example $x \mapsto \sum_{n=1}^{\infty}(1-2^{-n})x_n$ on ℓ_1. Therefore, if c-topologically surjective operators are defined via open balls as in definition 1.1, then 1-topological projectivity is an empty notion in* **Ban** *as well, and the free Banach spaces $\ell_1(\Lambda)$ are only $(1+\varepsilon)$-topologically projective for every $\varepsilon > 0$. The metric theory of [2] avoids this problem by posing lifting problems along strictly coisometric operators, that is, operators mapping the closed unit ball onto the closed unit ball; with this convention $\ell_1(\Lambda)$ and $\ell_1^0(\Lambda)$ are 1-topologically projective in* **Ban** *and* **Nor** *respectively, and 1-topological projectivity coincides with metric projectivity, as in [5, proposition 2.1.9]. For the purposes of this paper the choice of convention is immaterial: if we call $\xi : X \to Y$ strictly c-topologically surjective whenever $c\xi(B_X) \supset B_Y$ and define strict C-topological projectivity by lifting along such operators with the same bound $\|\psi\| \leq cC\|\phi\|$, then every C-topologically projective space is strictly C-topologically projective, and every strictly C-topologically projective space is $C(1+\varepsilon)$-topologically projective for every $\varepsilon > 0$. The first claim is obvious, since strictly c-topologically surjective operators are c-topologically surjective. For the second one, given a c-topologically surjective operator $\xi : X \to Y$ and $\varepsilon > 0$, endow X with the equivalent norm $\|x\|'' = \max(\|x\|, (1+\varepsilon)c\|\xi(x)\|)$; then $(1+\varepsilon)c\xi$ becomes a contractive operator mapping the closed unit ball onto the closed unit ball, and a lifting ψ'' along it with $\|\psi''\|'' \leq C\|\phi\|$ gives the lifting $\psi = (1+\varepsilon)c\psi''$ along ξ with $\|\psi\| \leq (1+\varepsilon)cC\|\phi\|$. In particular, the two conventions lead to the same notion of extremal projectivity.*

Our definition of extremal projectivity is equivalent to Helemskii's definition of extreme projectivity [3, definition I]: a normed space P is extremely projective if for every coisometric operator $\xi : X \to Y$, every bounded linear operator $\phi : P \to Y$ and every $\delta > 0$ there exists a lifting $\psi : P \to X$ of ϕ along ξ with $\|\psi\| < \|\phi\| + \delta$.

Proposition 2.4. *A normed space is extremally projective if and only if it is extremely projective in the sense of Helemskii.*

Proof. Let P be extremally projective, $\xi : X \to Y$ be a coisometric operator, $\phi : P \to Y$ be a bounded linear operator and $\delta > 0$. If $\phi = 0$ we take $\psi = 0$. Otherwise we put $C = 1 + \delta\|\phi\|^{-1}/2$. Since ξ is 1-topologically surjective, C-topological projectivity of P gives a lifting ψ of ϕ along ξ with $\|\psi\| \leq C\|\phi\| < \|\phi\| + \delta$.

Conversely, let P be extremely projective, $C > 1$, $\xi : X \to Y$ be a c-topologically surjective operator and $\phi : P \to Y$ be a non-zero bounded linear operator. Consider the operator $\xi' = c\xi$ and endow the vector space X with the norm $\|x\|' = \max(\|x\|, \|\xi'(x)\|)$, which is equivalent to the original one since $\|x\| \leq \|x\|' \leq \max(1, c\|\xi\|)\|x\|$. Denote the resulting normed space by X'. The operator $\xi' : X' \to Y$ is contractive by construction. If $y \in B_Y^\circ$, then by c-topological surjectivity of ξ there is $x \in B_X^\circ$ with $\xi'(x) = y$, and $\|x\|' = \max(\|x\|, \|y\|) < 1$. Thus, $\xi' : X' \to Y$ is coisometric. By extreme projectivity there is a lifting $\psi' : P \to X'$ of ϕ along ξ'

with $\|\psi'\| < \|\phi\| + (C-1)\|\phi\| = C\|\phi\|$. Regarded as an operator into X, the operator $\psi = c\psi'$ satisfies $\xi\psi = \xi'\psi' = \phi$ and $\|\psi\| \le c\|\psi'\|' \le cC\|\phi\|$. Since $C > 1$ is arbitrary, P is extremally projective. □

Next we record the standard permanence properties. The proofs are the same as in the Banach case, see [4, lemma 2.7] or [5, proposition 2.1.3], and we include them only for completeness; for the second part of the next proposition see also [3, proposition 1.1].

Proposition 2.5. *Let P' be a C-topologically projective normed space and P be a c-retract of P'. Then P is cC-topologically projective. Consequently, a normed space which is a $(1+\varepsilon)$-retract of an extremally projective normed space for every $\varepsilon > 0$ is extremally projective.*

Proof. Let $\zeta : P' \to P$ and $\eta : P \to P'$ be bounded linear operators with $\zeta\eta = 1_P$ and $\|\zeta\|\|\eta\| \le c$. Consider a c'-topologically surjective operator $\xi : X \to Y$ and a bounded linear operator $\phi : P \to Y$. By C-topological projectivity of P' there is a lifting $\psi' : P' \to X$ of $\phi\zeta$ along ξ with $\|\psi'\| \le c'C\|\phi\zeta\|$. Then $\psi = \psi'\eta$ satisfies $\xi\psi = \phi\zeta\eta = \phi$ and $\|\psi\| \le c'C\|\phi\|\|\zeta\|\|\eta\| \le c'cC\|\phi\|$. □

For a normed space P consider the operator

$$\pi_P : \ell_1^0(B_P) \to P, \quad e_x \mapsto x.$$

It is well defined and contractive, because the family $(x)_{x\in B_P}$ is bounded by 1. Moreover, π_P maps the closed unit ball of $\ell_1^0(B_P)$ onto B_P, since $\pi_P(e_x) = x$ for $x \in B_P$. Thus, π_P is strictly coisometric and, a fortiori, coisometric.

Proposition 2.6. *Let P be a normed space and $C \ge 1$. Then*

(i) if P is C-topologically projective, then π_P has a right inverse bounded linear operator of norm at most C;

(ii) if π_P has a right inverse bounded linear operator of norm at most C, then P is $C(1+\varepsilon)$-topologically projective for every $\varepsilon > 0$.

Proof. (*i*) Lift 1_P along the 1-topologically surjective operator π_P: we get $\sigma : P \to \ell_1^0(B_P)$ with $\pi_P\sigma = 1_P$ and $\|\sigma\| \le C$.

(*ii*) If $\pi_P\sigma = 1_P$ with $\|\sigma\| \le C$, then P is a C-retract of $\ell_1^0(B_P)$, which is $(1+\varepsilon)$-topologically projective by proposition 2.1. It remains to apply proposition 2.5. □

The equivalence of conditions (*i*) and (*iii*) in the following theorem is the main observation of [3, section 2] on extremally projective normed spaces: a normed space is extremally projective if and only if it is a near-retract of a free normed space, where a near-retract is a retract via a contractive operator which has right inverses of norm $< 1+\varepsilon$ for every $\varepsilon > 0$.

Theorem 2.7. *For a normed space P the following conditions are equivalent:*

(i) P is extremally projective;

(ii) for every $\varepsilon > 0$ the operator π_P has a right inverse bounded linear operator of norm at most $1+\varepsilon$;

(iii) for every $\varepsilon > 0$ there is a set Λ such that P is a $(1+\varepsilon)$-retract of $\ell_1^0(\Lambda)$;

(iv) for every $\varepsilon > 0$ there are a set Λ, a subspace $Z \subset \ell_1^0(\Lambda)$ and a projection $Q : \ell_1^0(\Lambda) \to \ell_1^0(\Lambda)$ onto Z with $\|Q\| \leq 1+\varepsilon$ such that $d_{BM}(P, Z) \leq 1+\varepsilon$.

Proof. $(i) \implies (ii)$ is proposition 2.6 (i), and $(ii) \implies (iii)$ is obvious.

$(iii) \implies (iv)$ Let $\zeta : \ell_1^0(\Lambda) \to P$ and $\eta : P \to \ell_1^0(\Lambda)$ satisfy $\zeta\eta = 1_P$ and $\|\zeta\|\|\eta\| \leq 1+\varepsilon$. Replacing ζ by $\|\zeta\|^{-1}\zeta$ and η by $\|\zeta\|\eta$ we may assume that $\|\zeta\| = 1$ and $\|\eta\| \leq 1+\varepsilon$. Put $Z = \eta(P)$ and $Q = \eta\zeta$. Then $Q^2 = \eta(\zeta\eta)\zeta = Q$, $\mathrm{Im}(Q) = Z$ and $\|Q\| \leq 1+\varepsilon$. The operator $\eta : P \to Z$ is a bijection with $\|\eta\| \leq 1+\varepsilon$ and $\eta^{-1} = \zeta|_Z$, so $\|\eta^{-1}\| \leq 1$ and $d_{BM}(P, Z) \leq 1+\varepsilon$.

$(iv) \implies (i)$ Let $T : P \to Z$ be a linear bijection with $\|T\|\|T^{-1}\| \leq 1+\varepsilon$. Put $\eta = T$ and $\zeta = T^{-1}Q : \ell_1^0(\Lambda) \to P$. Then $\zeta\eta = T^{-1}QT = T^{-1}T = 1_P$ and $\|\zeta\|\|\eta\| \leq (1+\varepsilon)^2$. By propositions 2.1 and 2.5 the space P is $(1+\varepsilon)^3$-topologically projective. As $\varepsilon > 0$ is arbitrary, P is extremally projective. □

We finish this section with a simple way of constructing new extremally projective spaces from old ones. For a family $(P_\lambda)_{\lambda\in\Lambda}$ of normed spaces we denote by $\bigoplus_1^0\{P_\lambda : \lambda \in \Lambda\}$ the subspace of the ℓ_1-sum $\bigoplus_1\{P_\lambda : \lambda \in \Lambda\}$ consisting of families with finitely many non-zero members. Clearly, $\bigoplus_1^0\{\mathbb{C} : \lambda \in \Lambda\} = \ell_1^0(\Lambda)$.

Proposition 2.8. *Let $(P_\lambda)_{\lambda\in\Lambda}$ be a family of extremally projective normed spaces. Then $\bigoplus_1^0\{P_\lambda : \lambda \in \Lambda\}$ is extremally projective. If Λ is finite, then $\bigoplus_1\{P_\lambda : \lambda \in \Lambda\}$ is extremally projective.*

Proof. Fix $\varepsilon > 0$. By theorem 2.7 for every $\lambda \in \Lambda$ there are a set S_λ and operators $\zeta_\lambda : \ell_1^0(S_\lambda) \to P_\lambda$, $\eta_\lambda : P_\lambda \to \ell_1^0(S_\lambda)$ with $\zeta_\lambda\eta_\lambda = 1_{P_\lambda}$, $\|\zeta_\lambda\| \leq 1$ and $\|\eta_\lambda\| \leq 1+\varepsilon$. Let S be the disjoint union of the sets S_λ. Then $\ell_1^0(S) = \bigoplus_1^0\{\ell_1^0(S_\lambda) : \lambda \in \Lambda\}$ isometrically, and the operators $\zeta = \bigoplus_1^0 \zeta_\lambda$ and $\eta = \bigoplus_1^0 \eta_\lambda$ are well defined bounded linear operators between $\ell_1^0(S)$ and $P = \bigoplus_1^0\{P_\lambda : \lambda \in \Lambda\}$ with $\zeta\eta = 1_P$, $\|\zeta\| \leq 1$ and $\|\eta\| \leq 1+\varepsilon$. By theorem 2.7 the space P is extremally projective. For finite Λ the sums $\bigoplus_1$ and $\bigoplus_1^0$ coincide. □

3 Dimension restrictions on topologically projective normed spaces

The free normed spaces $\ell_1^0(\Lambda)$ are far from being complete: for an infinite set Λ the Hamel basis $(e_\lambda)_{\lambda\in\Lambda}$ of $\ell_1^0(\Lambda)$ has the same cardinality as a dense subset of $\ell_1^0(\Lambda)$ of minimal cardinality, whereas the algebraic dimension of an infinite dimensional Banach space is uncountable by the Baire category theorem, and in fact at least continuum [10]. In this section we show that this smallness is inherited by all topologically projective normed spaces. Helemskii proved that a separable topologically projective normed space has at most countable algebraic dimension [3, proposition 2.4], and deduced that $\ell_1(\Lambda)$ is not topologically projective in **Nor** for any infinite set Λ [3, proposition 2.5]. His argument applies verbatim to separable subspaces of non-separable spaces, and in this hereditary form it will be convenient for us.

Theorem 3.1. *Let P be a topologically projective normed space. Then every separable subspace of P has at most countable algebraic dimension.*

Proof. By proposition 2.6 (i) there exists a bounded linear operator $\sigma : P \to \ell_1^0(B_P)$ with $\pi_P\sigma = 1_P$; in particular, σ is injective. Let Z be a separable subspace of P and M be a countable dense subset of Z. The set $S_0 = \bigcup_{x\in M} \mathrm{supp}(\sigma(x))$ is countable. If $z \in Z$, then $\sigma(z)$ is the limit in $\ell_1(B_P)$ of a sequence of elements of $\sigma(M)$, which are supported in S_0; since the coordinate functionals are

continuous, $\sigma(z)$ is supported in S_0 as well. Thus, $\sigma(Z) \subset \ell_1^0(S_0) = \operatorname{span}\{e_s : s \in S_0\}$, a space of countable algebraic dimension, and $\sigma|_Z$ is injective. □

Corollary 3.2. *Let P be a topologically projective normed space, Z be a subspace of P and $\|\cdot\|'$ be a norm on Z such that $(Z, \|\cdot\|')$ is complete and $\|z\| \le K\|z\|'$ for all $z \in Z$ and some constant $K > 0$. Then Z is finite dimensional. In particular, every complete subspace of P is finite dimensional, and no infinite dimensional Banach space is topologically projective in* **Nor**.

Proof. Assume that Z is infinite dimensional. Choose a linearly independent sequence $(z_n)_{n\in\mathbb{N}}$ in Z and let Z_0 be the closure of its linear span in $(Z, \|\cdot\|')$. Then $(Z_0, \|\cdot\|')$ is a separable infinite dimensional Banach space, so its algebraic dimension is uncountable. On the other hand, Z_0 is separable with respect to $\|\cdot\|$ because $\|\cdot\| \le K\|\cdot\|'$, so by theorem 3.1 the algebraic dimension of Z_0 is countable. This contradiction proves the first claim; the rest follows by taking $Z = P$ and $\|\cdot\|' = \|\cdot\|$. □

Let us compare theorem 3.1 with the theorem of Grønbæk [9, theorem 2.1]: a normed space E is topologically projective if and only if it is topologically isomorphic to $\ell_1^0(\Lambda)$, where the cardinality of Λ is the density character of E. Since for infinite Λ the algebraic dimension of $\ell_1^0(\Lambda)$ equals the cardinality of Λ, the algebraic dimension of an infinite dimensional topologically projective normed space coincides with its density character. Theorem 3.1 is a hereditary version of this equality: it applies to all separable subspaces of P, which are, of course, not topologically projective in general.

The following example shows that corollary 3.2 is sharper than its last claim: a normed space may have no infinite dimensional complete subspaces and still fail to be topologically projective. It also shows that topological projectivity of a normed space is not determined by its completion.

Example 3.3. *Let $(w_n)_{n\in\mathbb{N}}$ be a sequence of real numbers with $w_n \ge 1$ and $\lim_n w_n = +\infty$, and let*

$$P_w = \left\{ x \in \ell_1 : \sum_{n=1}^{\infty} w_n |x_n| < +\infty \right\}$$

with the norm inherited from ℓ_1. Then P_w is a dense subspace of ℓ_1 containing $\ell_1^0(\mathbb{N})$, so its completion is ℓ_1, and

(i) every complete subspace of P_w is finite dimensional;

(ii) P_w is not topologically projective.

Proof. Denote $\|x\|_w = \sum_{n=1}^{\infty} w_n|x_n|$ for $x \in P_w$. Then $(P_w, \|\cdot\|_w)$ is the Banach space $\ell_1(w)$ of w-weighted summable sequences and $\|x\| \le \|x\|_w$ for $x \in P_w$.

(i) Let Z be a complete subspace of P_w. If $(z_k)_{k\in\mathbb{N}}$ is a sequence in Z converging to $z \in P_w$ with respect to $\|\cdot\|_w$, then it converges to z in ℓ_1, so $z \in Z$. Thus, Z is closed in $(P_w, \|\cdot\|_w)$ and the two norms $\|\cdot\|$ and $\|\cdot\|_w$ are complete on Z. Since $\|\cdot\| \le \|\cdot\|_w$, by the open mapping theorem there is a constant K with $\|z\|_w \le K\|z\|$ for all $z \in Z$. Assume that Z is infinite dimensional and fix $N \in \mathbb{N}$. The linear map $Z \to \mathbb{C}^N$, $z \mapsto (z_1, \dots, z_N)$ has a non-trivial kernel, so there exists $z \in Z$ with $\|z\| = 1$ and $z_n = 0$ for $n \le N$. Then $K \ge \|z\|_w = \sum_{n>N} w_n|z_n| \ge \inf_{n>N} w_n$. Since N is arbitrary and $\lim_n w_n = +\infty$, we get a contradiction.

(ii) Apply corollary 3.2 to $Z = P_w$ and $\|\cdot\|' = \|\cdot\|_w$. Alternatively, note that P_w is separable, being dense in ℓ_1, while its algebraic dimension is uncountable, since $(P_w, \|\cdot\|_w)$ is an infinite dimensional Banach space; so theorem 3.1 applies directly. □

4 Extremally projective spaces which are not free

In this section we show that extremal projectivity is not an isometric invariant of the form "P is isometrically isomorphic to $\ell_1^0(\Lambda)$". Our examples are subspaces of $\ell_1 = \ell_1(\mathbb{N})$ containing $\ell_1^0(\mathbb{N})$. For a sequence $x \in \ell_1$ and $N \in \mathbb{N} \cup \{0\}$ we denote by $x|_N$ the sequence which coincides with x at $n \le N$ and vanishes at $n > N$. For a finite dimensional subspace V of ℓ_1 we consider the normed space

$$P_V = \ell_1^0(\mathbb{N}) + V \subset \ell_1$$

with the norm inherited from ℓ_1. Clearly, $\ell_1^0(\mathbb{N}) \subset P_V \subset \ell_1$, so P_V is dense in ℓ_1. Since $\ell_1^0(\mathbb{N})$ has countable algebraic dimension, so does P_V.

Theorem 4.1. *Let $v \in \ell_1$, $N \in \mathbb{N} \cup \{0\}$ and $0 < \delta < 1$. Then there exists a bounded linear operator $R : \ell_1 \to \ell_1$ with $\|R\| \le \delta$ such that*

(i) $R(x) = 0$ for every $x \in \ell_1$ with $x = x|_N$;

(ii) $R(\ell_1^0(\mathbb{N})) \subset \ell_1^0(\mathbb{N})$;

(iii) $(1_{\ell_1} - R)(v) \in \ell_1^0(\mathbb{N})$;

(iv) $(1_{\ell_1} - R)^{-1}(\ell_1^0(\mathbb{N})) \subset \ell_1^0(\mathbb{N}) + \mathbb{C}v$.

In particular, $T = 1_{\ell_1} - R$ is a linear bijection of ℓ_1 with $\|T\| \le 1 + \delta$ and $\|T^{-1}\| \le (1-\delta)^{-1}$ such that $T(x) = x$ whenever $x = x|_N$, $T(\ell_1^0(\mathbb{N})) \subset \ell_1^0(\mathbb{N})$ and $T(\ell_1^0(\mathbb{N}) + \mathbb{C}v) = \ell_1^0(\mathbb{N})$.

Proof. Since $\|R\| \le \delta < 1$, the operator $T = 1_{\ell_1} - R$ is a linear bijection with $T^{-1} = \sum_{m=0}^{\infty} R^m$, $\|T\| \le 1 + \delta$ and $\|T^{-1}\| \le (1-\delta)^{-1}$, and the properties of T listed in the last sentence follow from (i)–(iv). So we only need to construct R with the properties (i)–(iv).

If $v - v|_N$ is finitely supported, take $R = 0$. Assume that $v - v|_N$ has infinite support. For $M \ge N$ put $t_M = \|v - v|_M\| = \sum_{n>M} |v_n|$. Then the sequence $(t_M)_{M \ge N}$ is non-increasing, $t_M > 0$ for all $M \ge N$ because $v - v|_N$ has infinite support, and $\lim_M t_M = 0$. Choose integers $N = N_0 < N_1 < N_2 < \dots$ such that $t_{N_k} \le (\delta/2) t_{N_{k-1}}$ for all $k \in \mathbb{N}$. The sets $B_k = \{n \in \mathbb{N} : N_{k-1} < n \le N_k\}$, $k \in \mathbb{N}$, are finite, pairwise disjoint and cover $\{n \in \mathbb{N} : n > N\}$. Put $w_k = v|_{N_k} - v|_{N_{k-1}}$, that is, w_k is the restriction of v to B_k. Then $w_k \in \ell_1^0(\mathbb{N})$, $v - v|_N = \sum_{k=1}^{\infty} w_k$ in ℓ_1, $\operatorname{supp}(w_k) \subset B_k$, and

$$\|w_k\| = t_{N_{k-1}} - t_{N_k} \ge (1 - \delta/2) t_{N_{k-1}} > 0, \qquad \|w_{k+1}\| \le t_{N_k} \le (\delta/2) t_{N_{k-1}} \le \frac{\delta/2}{1-\delta/2} \|w_k\| \le \delta \|w_k\|,$$

where the last inequality holds because $\delta \le 1$. For $k \in \mathbb{N}$ define the functional

$$g_k : \ell_1 \to \mathbb{C}, \quad g_k(x) = \sum_{n \in B_k} \overline{\operatorname{sgn}(v_n)}\, x_n,$$

where $\operatorname{sgn}(z) = z/|z|$ for $z \ne 0$ and $\operatorname{sgn}(0) = 0$. Then $|g_k(x)| \le \sum_{n \in B_k} |x_n|$ for all $x \in \ell_1$, $g_k(w_k) = \|w_k\|$ and $g_k(w_j) = 0$ for $j \ne k$, because the supports of w_j and g_k are disjoint. Now define

$$R : \ell_1 \to \ell_1, \quad R(x) = \sum_{k=1}^{\infty} \frac{g_k(x)}{\|w_k\|}\, w_{k+1}.$$

The series converges absolutely, since

$$\sum_{k=1}^{\infty} \frac{|g_k(x)|}{\|w_k\|}\|w_{k+1}\| \le \delta \sum_{k=1}^{\infty}\sum_{n\in B_k}|x_n| = \delta\|x\|,$$

so R is a well defined bounded linear operator with $\|R\| \le \delta$. We verify the properties (i)–(iv).

(i) If $x = x|_N$, then $g_k(x) = 0$ for all k, because $\operatorname{supp}(g_k) \subset B_k \subset \{n : n > N\}$. Hence $R(x) = 0$.

(ii) If $x \in \ell_1^0(\mathbb{N})$, then $g_k(x) \neq 0$ only for the finitely many k with $B_k \cap \operatorname{supp}(x) \neq \varnothing$, so $R(x)$ is a finite linear combination of the finitely supported sequences w_{k+1}.

(iii) Since $g_k(v|_N) = 0$ and $g_k(v - v|_N) = \sum_{j=1}^{\infty} g_k(w_j) = g_k(w_k) = \|w_k\|$, we get $R(v) = \sum_{k=1}^{\infty} w_{k+1} = v - v|_N - w_1$, so $(1_{\ell_1} - R)(v) = v|_N + w_1 \in \ell_1^0(\mathbb{N})$.

(iv) Let $x \in \ell_1^0(\mathbb{N})$ and let K be the finite set of those k for which $g_k(x) \neq 0$. Put $c_k = g_k(x)/\|w_k\|$ for $k \in K$. Since $g_j(w_k) = \delta_{jk}\|w_k\|$, we have $R(w_k) = w_{k+1}$ for all k, hence $R^m(x) = \sum_{k\in K} c_k w_{k+m}$ for $m \ge 1$ and

$$(1_{\ell_1} - R)^{-1}(x) = \sum_{m=0}^{\infty} R^m(x) = x + \sum_{k\in K} c_k \sum_{m=1}^{\infty} w_{k+m} = x + \sum_{k\in K} c_k(v - v|_{N_k}) \in \ell_1^0(\mathbb{N}) + \mathbb{C}v.$$

□

Theorem 4.2. *Let V be a finite dimensional subspace of ℓ_1. Then $d_{BM}(P_V, \ell_1^0(\mathbb{N})) = 1$. Consequently, P_V is extremally projective.*

Proof. We use induction on $m = \dim V$. If $m = 0$, then $P_V = \ell_1^0(\mathbb{N})$. Let $m \ge 1$, fix $0 < \delta < 1$ and write $V = \mathbb{C}v + V'$ with $\dim V' = m - 1$. Let T be the operator given by theorem 4.1 for v, $N = 0$ and δ. Then

$$T(P_V) = T(\ell_1^0(\mathbb{N}) + \mathbb{C}v) + T(V') = \ell_1^0(\mathbb{N}) + T(V') = P_{T(V')},$$

and $T(V')$ is a subspace of ℓ_1 of dimension $m-1$. The restriction of T to P_V is a linear bijection onto $P_{T(V')}$ with $\|T|_{P_V}\|\|(T|_{P_V})^{-1}\| \le (1+\delta)/(1-\delta)$, so $d_{BM}(P_V, \ell_1^0(\mathbb{N})) \le \frac{1+\delta}{1-\delta} d_{BM}(P_{T(V')}, \ell_1^0(\mathbb{N})) = \frac{1+\delta}{1-\delta}$ by the induction hypothesis. Since δ is arbitrary, $d_{BM}(P_V, \ell_1^0(\mathbb{N})) = 1$.

Now let $\varepsilon > 0$ and let $S : P_V \to \ell_1^0(\mathbb{N})$ be a linear bijection with $\|S\|\|S^{-1}\| \le 1 + \varepsilon$. Then P_V is a $(1+\varepsilon)$-retract of $\ell_1^0(\mathbb{N})$ via S^{-1} and S, and by propositions 2.1 and 2.5 it is $(1+\varepsilon)^2$-topologically projective. As ε is arbitrary, P_V is extremally projective. □

To see that the spaces P_V with $V \not\subset \ell_1^0(\mathbb{N})$ are not isometrically isomorphic to free normed spaces we look at extreme points of unit balls. By $\operatorname{ext}(B)$ we denote the set of extreme points of a convex set B and by $\operatorname{conv}(M)$ the convex hull of a set M.

Lemma 4.3. *Let S be a set and E be a subspace of $\ell_1(S)$ containing $\ell_1^0(S)$, endowed with the norm of $\ell_1(S)$. Then $\operatorname{ext}(B_E) = \{\lambda e_s : s \in S,\ \lambda \in \mathbb{C},\ |\lambda| = 1\}$ and $\operatorname{conv}(\operatorname{ext}(B_E)) = B_{\ell_1^0(S)}$.*

Proof. The points λe_s with $|\lambda| = 1$ are extreme points of $B_{\ell_1(S)}$, and $B_E = B_{\ell_1(S)} \cap E$, so they are extreme points of B_E. Conversely, let $x \in B_E$ be not of this form. If $x = 0$, then $x = \frac{1}{2}e_s + \frac{1}{2}(-e_s)$ for any $s \in S$. If $0 < \|x\| < 1$, then $x = \frac{1}{2}(1+t)x + \frac{1}{2}(1-t)x$ for $t = 1 - \|x\| > 0$, and both $(1 \pm t)x$

belong to B_E. If $\|x\| = 1$, then x has at least two non-zero coordinates x_n and x_m, $n \neq m$. Put $t = \min(|x_n|, |x_m|) > 0$ and $u = t(\operatorname{sgn}(x_n)e_n - \operatorname{sgn}(x_m)e_m) \in \ell_1^0(S) \subset E$. Then $x \pm u \in E$ and

$$\|x \pm u\| = (|x_n| \pm t) + (|x_m| \mp t) + \sum_{s \neq n,m} |x_s| = \|x\| = 1,$$

so $x = \frac{1}{2}(x+u) + \frac{1}{2}(x-u)$ with $u \neq 0$. In all three cases x is not an extreme point of B_E.

Every finite convex combination of the points λe_s is a finitely supported element of $B_{\ell_1(S)}$, so $\operatorname{conv}(\operatorname{ext}(B_E)) \subset B_{\ell_1^0(S)}$. Conversely, if $x \in B_{\ell_1^0(S)}$, then

$$x = \sum_{s \in \operatorname{supp}(x)} |x_s| \operatorname{sgn}(x_s) e_s + (1 - \|x\|) \cdot 0,$$

and $0 = \frac{1}{2}e_s + \frac{1}{2}(-e_s)$, so $x \in \operatorname{conv}(\operatorname{ext}(B_E))$. □

Theorem 4.4. *Let S be a set and E be a subspace of $\ell_1(S)$ containing $\ell_1^0(S)$, endowed with the norm of $\ell_1(S)$. If E is isometrically isomorphic to $\ell_1^0(\Lambda)$ for some set Λ, then $E = \ell_1^0(S)$.*

Proof. Let $T : \ell_1^0(\Lambda) \to E$ be an isometric isomorphism. Then $T(B_{\ell_1^0(\Lambda)}) = B_E$ and T maps $\operatorname{ext}(B_{\ell_1^0(\Lambda)})$ onto $\operatorname{ext}(B_E)$. Applying lemma 4.3 to E and to $\ell_1^0(\Lambda)$ itself we get

$$B_E = T(B_{\ell_1^0(\Lambda)}) = T(\operatorname{conv}(\operatorname{ext}(B_{\ell_1^0(\Lambda)}))) = \operatorname{conv}(T(\operatorname{ext}(B_{\ell_1^0(\Lambda)}))) = \operatorname{conv}(\operatorname{ext}(B_E)) = B_{\ell_1^0(S)}.$$

Hence $E = \ell_1^0(S)$. □

Corollary 4.5. *Let V be a finite dimensional subspace of ℓ_1 with $V \not\subset \ell_1^0(\mathbb{N})$, for example $V = \mathbb{C}v$ for $v = (2^{-n})_{n \in \mathbb{N}}$. Then P_V is an extremally projective normed space which is not isometrically isomorphic to $\ell_1^0(\Lambda)$ for any set Λ.*

Proof. Combine theorems 4.2 and 4.4. □

Corollary 4.5 answers the question of Helemskii from [3, section 2] and [2, section 1] in the negative and disproves [9, conjecture 3.1]. Note that by theorem 4.4 every subspace E with $\ell_1^0(\mathbb{N}) \subsetneq E \subset \ell_1$ fails to be isometrically isomorphic to a free normed space, but by example 3.3 not all of these subspaces are extremally projective. Thus, between $\ell_1^0(\mathbb{N})$ and ℓ_1 there are both extremally projective and non-projective normed spaces, and the difference between them is invisible on the level of completions. The next section provides a criterion which distinguishes them.

5 Extremally projective normed spaces of countable dimension

We start with a necessary condition which holds in all dimensions. Recall that the completion $\overline{P}$ of a normed space P is a Banach space containing P as a dense subspace. Every bounded linear operator between normed spaces extends uniquely to a bounded linear operator between their completions with the same norm, and these extensions respect compositions. The following theorem is [3, proposition 2.3], see also [9, section 3]; we include the proof to make the paper self-contained.

Theorem 5.1. *Let P be an extremally projective normed space. Then $\overline{P}$ is isometrically isomorphic to $\ell_1(\Gamma)$ for some set Γ.*

Proof. Fix $\varepsilon > 0$. By theorem 2.7 there is a bounded linear operator $\sigma : P \to \ell_1^0(B_P)$ with $\pi_P\sigma = 1_P$ and $\|\sigma\| \le 1+\varepsilon$. The completion of $\ell_1^0(B_P)$ is $\ell_1(B_P)$, so π_P and σ extend to bounded linear operators $\overline{\pi} : \ell_1(B_P) \to \overline{P}$ and $\overline{\sigma} : \overline{P} \to \ell_1(B_P)$ with $\overline{\pi}\,\overline{\sigma} = 1_{\overline{P}}$, $\|\overline{\pi}\| \le 1$ and $\|\overline{\sigma}\| \le 1+\varepsilon$. Thus, $\overline{P}$ is a $(1+\varepsilon)$-retract of $\ell_1(B_P)$ in **Ban**. The proof of proposition 2.1 shows that $\ell_1(\Lambda)$ is $(1+\varepsilon)$-topologically projective in **Ban** for every set Λ: in the notation of that proof, for a Banach space X the family $(x_\lambda)_{\lambda\in\Lambda}$ defines a bounded linear operator $\psi : \ell_1(\Lambda) \to X$, $\psi(x) = \sum_{\lambda\in\Lambda} x(\lambda)x_\lambda$, where the series converges absolutely. The proof of proposition 2.5 works in **Ban** verbatim, so $\overline{P}$ is $(1+\varepsilon)^2$-topologically projective in **Ban**. Since ε is arbitrary, $\overline{P}$ is extremally projective in **Ban**. The proof of proposition 2.4 also works in **Ban** verbatim, because the space X' constructed there is complete whenever X is. Hence $\overline{P}$ is extremely projective in **Ban** in the sense of Helemskii, and by Grothendieck's theorem in the form given in [3, section 2] (see also [7] and [2, proposition 3.2]) $\overline{P}$ is isometrically isomorphic to $\ell_1(\Gamma)$ for some set Γ.

For the convenience of the reader we sketch the classical argument behind the last step. Taking adjoints of $\overline{\pi}$ and $\overline{\sigma}$ we see that $\overline{P}^*$ is a $(1+\varepsilon)$-retract of the 1-injective Banach space $\ell_\infty(B_P)$, so $\overline{P}^*$ is $(1+\varepsilon)$-injective for every $\varepsilon > 0$, that is, every bounded linear operator from a subspace of a Banach space into $\overline{P}^*$ extends to the whole space with norm increased by at most $1+\varepsilon$. Passing to the infimum over ε we get that $\overline{P}^*$ is 1-injective, hence isometrically isomorphic to $C(K)$ for a Stonean space K [8, theorem 3.11.6], and by Grothendieck's theorem [7] (see [8, section 17]) a Banach space whose dual is 1-injective is isometrically isomorphic to $L_1(\Omega,\mu)$ for some measure space (Ω,μ). Finally, $\overline{P}$ is isomorphic to a complemented subspace of $\ell_1(B_P)$ and therefore has the Schur property, which forces μ to be purely atomic, and L_1 of a purely atomic measure is isometrically isomorphic to $\ell_1(\Gamma)$, where Γ is the set of atoms. □

Corollary 5.2. *A finite dimensional normed space is extremally projective if and only if it is isometrically isomorphic to ℓ_1^n for some $n \in \mathbb{N} \cup \{0\}$.*

Proof. A finite dimensional normed space coincides with its completion, so the necessity follows from theorem 5.1. The sufficiency follows from proposition 2.1, since $\ell_1^n = \ell_1^0(\{1,\dots,n\})$. □

Thus, the examples of section 4 are necessarily infinite dimensional. Theorems 3.1 and 5.1 describe extremally projective normed spaces from two opposite sides: they are dense subspaces of the spaces $\ell_1(\Gamma)$ which are small enough to have separable subspaces of countable algebraic dimension only. In this section we show that for spaces of countable dimension there is nothing in between: every dense subspace of ℓ_1 of countable algebraic dimension is extremally projective, and even almost isometric to $\ell_1^0(\mathbb{N})$. This answers the question of Helemskii [3, section 2] on dense subspaces of $\ell_1(\Lambda)$ for countable Λ. The proof is a back-and-forth construction based on theorem 4.1 and on the following two lemmas.

Lemma 5.3. *Let Y be a finite dimensional subspace of $\ell_1^0(\mathbb{N})$ and $n \in \mathbb{N}$ be such that $e_n \notin Y$. Then there is a constant $C \ge 1$ such that for every $z \in \ell_1^0(\mathbb{N})$ with $\|z - e_n\| \le (4C)^{-1}$ there exists a linear bijection $S : \ell_1 \to \ell_1$ with $\|S - 1_{\ell_1}\| \le 2C\|z - e_n\| \le 1/2$, $S(y) = y$ for all $y \in Y$ and $S(z) = e_n$.*

Proof. Since Y is finite dimensional, it is closed in ℓ_1, so $d = \inf\{\|e_n - y\| : y \in Y\} > 0$. By the Hahn-Banach theorem there is $f \in \ell_1^*$ with $f(y) = 0$ for $y \in Y$, $f(e_n) = 1$ and $\|f\| = d^{-1}$. Put $C = \max(1, d^{-1})$. Let $z \in \ell_1^0(\mathbb{N})$ satisfy $\|z - e_n\| \le (4C)^{-1}$. Then $|f(z) - 1| = |f(z - e_n)| \le C\|z - e_n\| \le 1/4$, so $|f(z)| \ge 3/4$. Define

$$S : \ell_1 \to \ell_1, \quad S(x) = x + \frac{f(x)}{f(z)}(e_n - z).$$

Then $S(y) = y$ for $y \in Y$ because f vanishes on Y, and $S(z) = z + (e_n - z) = e_n$. Moreover, $\|S - 1_{\ell_1}\| \le \|f\||f(z)|^{-1}\|e_n - z\| \le \frac{4}{3}C\|z - e_n\| \le 2C\|z - e_n\| \le 1/2$, so S is a linear bijection of ℓ_1 by the Neumann series. □

Lemma 5.4. *Let X be a Banach space and $(F_m)_{m\in\mathbb{N}}$ be a sequence of bounded linear operators on X such that $\theta_m = \|F_m - 1_X\| < 1$ for all m and $\sum_{m=1}^{\infty}\theta_m < +\infty$. Put $A_j = F_jF_{j-1}\cdots F_1$ for $j \in \mathbb{N}$. Then the sequence $(A_j)_{j\in\mathbb{N}}$ converges in the operator norm to a linear bijection T of X with*

$$\|T\| \le \prod_{m=1}^{\infty}(1+\theta_m), \qquad \|T^{-1}\| \le \prod_{m=1}^{\infty}(1-\theta_m)^{-1}.$$

Proof. Both products converge because $\sum_m \theta_m < +\infty$; denote them by M and M'. Then $\|A_j\| \le M$ and, since each F_m is invertible with $\|F_m^{-1}\| \le (1-\theta_m)^{-1}$ by the Neumann series, $\|A_j^{-1}\| \le M'$ for all j. From $\|A_j - A_{j-1}\| = \|(F_j - 1_X)A_{j-1}\| \le \theta_j M$ we see that $(A_j)_j$ is a Cauchy sequence; let T be its limit, so $\|T\| \le M$. Also $\|A_j^{-1} - A_k^{-1}\| = \|A_j^{-1}(A_k - A_j)A_k^{-1}\| \le M'^2\|A_k - A_j\|$, so $(A_j^{-1})_j$ is a Cauchy sequence with a limit B, $\|B\| \le M'$. Passing to the limit in $A_jA_j^{-1} = A_j^{-1}A_j = 1_X$ we get $TB = BT = 1_X$. □

Theorem 5.5. *Let D be a dense subspace of ℓ_1 of countable algebraic dimension and $\varepsilon > 0$. Then there exists a linear bijection $T : \ell_1 \to \ell_1$ with $\|T\|\|T^{-1}\| \le 1+\varepsilon$ and $T(D) = \ell_1^0(\mathbb{N})$. In particular, $d_{BM}(D, \ell_1^0(\mathbb{N})) = 1$.*

Proof. Since ℓ_1 is infinite dimensional and D is dense in ℓ_1, the algebraic dimension of D is countably infinite. Let $(d_j)_{j\in\mathbb{N}}$ be a Hamel basis of D. Choose a sequence $(\theta_m)_{m\in\mathbb{N}}$ of numbers in $(0, 1/2]$ with $\prod_{m=1}^{\infty}(1+\theta_m)(1-\theta_m)^{-1} \le 1+\varepsilon$.

We construct by induction linear bijections $F_1, F_2, \ldots$ of ℓ_1 with $\|F_m - 1_{\ell_1}\| \le \theta_m$ and finite dimensional subspaces $Y_0 = \{0\} \subset Y_1 \subset Y_2 \subset \ldots$ of $\ell_1^0(\mathbb{N})$ such that, with $A_0 = 1_{\ell_1}$ and $A_j = F_{2j}F_{2j-1}A_{j-1}$, for every $j \in \mathbb{N}$ the following conditions hold:

(a) $A_j(d_i) \in Y_j$ for $i \le j$;

(b) $e_i \in Y_j$ for $i \le j$;

(c) $Y_j \subset A_j(D)$;

(d) F_{2j-1} and F_{2j} act as the identity on Y_{j-1}.

Suppose that A_{j-1} and Y_{j-1} have been constructed (for $j = 1$ there is nothing to suppose). Let N be an integer not smaller than every element of the supports of the vectors of some basis of Y_{j-1}, so that $y = y|_N$ for all $y \in Y_{j-1}$. Apply theorem 4.1 to the vector $v = A_{j-1}(d_j)$, the integer N and $\delta = \theta_{2j-1}$, and let $F_{2j-1} = 1_{\ell_1} - R$ be the resulting operator. Then $\|F_{2j-1} - 1_{\ell_1}\| \le \theta_{2j-1}$, F_{2j-1} acts as the identity on Y_{j-1}, and $y_j = F_{2j-1}A_{j-1}(d_j) \in \ell_1^0(\mathbb{N})$. Put $A' = F_{2j-1}A_{j-1}$ and $Y' = Y_{j-1} + \mathbb{C}y_j \subset \ell_1^0(\mathbb{N})$. Note that $Y' \subset A'(D)$: indeed, $Y_{j-1} \subset A_{j-1}(D)$ by the induction hypothesis (for $j = 1$ trivially), F_{2j-1} fixes Y_{j-1}, so $Y_{j-1} = F_{2j-1}(Y_{j-1}) \subset A'(D)$, and $y_j = A'(d_j) \in A'(D)$.

If $e_j \in Y'$, put $F_{2j} = 1_{\ell_1}$ and $Y_j = Y'$. Otherwise, let C be the constant given by lemma 5.3 for Y' and $n = j$, and let N' be an integer not smaller than j and than every element of the supports of the vectors of some basis of Y'. Since A' is a linear bijection of ℓ_1 and D is dense, $A'(D)$ is dense in ℓ_1, so there is $d \in D$ with

$$\|A'(d) - e_j\| \le \frac{\theta_{2j}}{16C}.$$

Apply theorem 4.1 to the vector $A'(d)$, the integer N' and $\delta = \theta_{2j}/2$, and let $G = 1_{\ell_1} - R'$ be the resulting operator. Then G fixes Y' and e_j, $\|G - 1_{\ell_1}\| \le \theta_{2j}/2$, $\|G\| \le 5/4$ because $\theta_{2j} \le 1/2$, and $z = GA'(d) \in \ell_1^0(\mathbb{N})$ satisfies

$$\|z - e_j\| = \|G(A'(d) - e_j)\| \le \frac{5\theta_{2j}}{64C} \le \frac{1}{4C}.$$

Let S be the operator given by lemma 5.3 for Y', $n = j$ and z. Then S fixes Y', $S(z) = e_j$ and $\|S - 1_{\ell_1}\| \le 2C\|z - e_j\| \le 5\theta_{2j}/32$. Put $F_{2j} = SG$ and $Y_j = Y' + \mathbb{C}e_j$. Then

$$\|F_{2j} - 1_{\ell_1}\| \le \|(S - 1_{\ell_1})G\| + \|G - 1_{\ell_1}\| \le \frac{5\theta_{2j}}{32} \cdot \frac{5}{4} + \frac{\theta_{2j}}{2} \le \theta_{2j}.$$

Let us check (a)–(d) for j. The operator F_{2j} acts as the identity on $Y' \supset Y_{j-1}$, which together with the first step gives (d). For $i < j$ we have $A_{j-1}(d_i) \in Y_{j-1}$, so $A_j(d_i) = F_{2j}F_{2j-1}A_{j-1}(d_i) = A_{j-1}(d_i) \in Y_j$, and $A_j(d_j) = F_{2j}(y_j) = y_j \in Y_j$; this is (a). Condition (b) holds by the definition of Y_j and the induction hypothesis. Finally, $Y' \subset A'(D)$ and F_{2j} fixes Y', so $Y' \subset F_{2j}A'(D) = A_j(D)$, and $e_j = SGA'(d) = A_j(d) \in A_j(D)$; this is (c).

By lemma 5.4 the operators A_j converge in norm to a linear bijection T of ℓ_1 with $\|T\|\|T^{-1}\| \le \prod_m (1+\theta_m)(1-\theta_m)^{-1} \le 1+\varepsilon$. For every $i \in \mathbb{N}$ and $j \ge i$ we have $A_j(d_i) \in Y_j$ by (a), and by (d) the operators $F_{2j+1}, F_{2j+2}, \dots$ fix Y_j, so $A_k(d_i) = A_j(d_i)$ for all $k \ge j$ and $T(d_i) = A_i(d_i) \in Y_i \subset \ell_1^0(\mathbb{N})$. Hence $T(D) \subset \ell_1^0(\mathbb{N})$. Similarly, for every $i \in \mathbb{N}$ by (b) and (c) there is $d \in D$ with $A_i(d) = e_i \in Y_i$, and by (d) we get $T(d) = A_i(d) = e_i$. Hence $\ell_1^0(\mathbb{N}) \subset T(D)$. □

Now we are ready to prove the main result of the paper.

Theorem 5.6. *Let P be a normed space of countable (finite or countably infinite) algebraic dimension. Then the following conditions are equivalent:*

(i) P is extremally projective;

(ii) $\overline{P}$ is isometrically isomorphic to $\ell_1(\Gamma)$ for some countable set Γ;

(iii) $d_{BM}(P, \ell_1^0(\Gamma)) = 1$ for some countable set Γ;

(iv) for every $\varepsilon > 0$ there is a set Λ with $d_{BM}(P, \ell_1^0(\Lambda)) < 1 + \varepsilon$.

Proof. $(i) \implies (ii)$ By theorem 5.1 the space $\overline{P}$ is isometrically isomorphic to $\ell_1(\Gamma)$ for some set Γ. The rational linear span of a countable Hamel basis of P is a countable dense subset of $\overline{P}$, so $\overline{P}$ is separable and Γ is countable.

$(ii) \implies (iii)$ Let $J : \overline{P} \to \ell_1(\Gamma)$ be an isometric isomorphism. If Γ is finite, then $\overline{P}$ is finite dimensional, so $P = \overline{P}$ and J maps P onto $\ell_1(\Gamma) = \ell_1^0(\Gamma)$. If Γ is infinite, we may assume that $\Gamma = \mathbb{N}$. Then $D = J(P)$ is a dense subspace of ℓ_1 of countable algebraic dimension, and by theorem 5.5 $d_{BM}(P, \ell_1^0(\mathbb{N})) = d_{BM}(D, \ell_1^0(\mathbb{N})) = 1$.

$(iii) \implies (iv)$ is obvious.

$(iv) \implies (i)$ Fix $\varepsilon > 0$ and a linear bijection $S : P \to \ell_1^0(\Lambda)$ with $\|S\|\|S^{-1}\| < 1 + \varepsilon$. Then P is a $(1+\varepsilon)$-retract of $\ell_1^0(\Lambda)$ via S^{-1} and S, so P is extremally projective by theorem 2.7. □

Corollary 5.7. *Every subspace of ℓ_1 of countable algebraic dimension which is dense in ℓ_1 is extremally projective and almost isometric to $\ell_1^0(\mathbb{N})$, that is, has Banach-Mazur distance 1 to $\ell_1^0(\mathbb{N})$.*

Combining theorem 5.6 with theorem 3.1 we obtain a complete description of separable extremally projective normed spaces.

Theorem 5.8. *Let P be a separable normed space. Then the following conditions are equivalent:*

(i) P is extremally projective;

(ii) P has at most countable algebraic dimension and $\overline{P}$ is isometrically isomorphic to $\ell_1(\Gamma)$ for some countable set Γ;

(iii) P has at most countable algebraic dimension and $d_{BM}(P, \ell_1^0(\Gamma)) = 1$ for some countable set Γ.

Proof. If P is extremally projective, then it has at most countable algebraic dimension by theorem 3.1 (or by [3, proposition 2.4]). It remains to apply theorem 5.6. □

Condition *(ii)* of theorem 5.6 shows that for normed spaces of countable dimension extremal projectivity is a property of the completion. By example 3.3 this is no longer true for spaces of uncountable dimension, and theorem 5.8 explains why: the dimension restriction of theorem 3.1 is an independent necessary condition. Note that example 3.3 is of the form $\ell_1^0(\mathbb{N}) + V$, as the extremally projective spaces of theorem 4.2 and corollary 5.7, but now V has uncountable dimension. On the other hand, uncountable dimension alone is not an obstruction.

Proposition 5.9. *Let Γ be a set and V be a subspace of $\ell_1(\Gamma)$ of countable algebraic dimension. Then $\ell_1^0(\Gamma) + V$ is extremally projective.*

Proof. The set $\Gamma_0 = \bigcup\{\mathrm{supp}(v) : v \in V\}$ is countable, being the union of the supports of the vectors of a countable Hamel basis of V. Put $P_0 = \ell_1^0(\Gamma_0) + V \subset \ell_1(\Gamma_0)$. Then $\ell_1^0(\Gamma) + V = \ell_1^0(\Gamma \setminus \Gamma_0) \bigoplus_1 P_0$ isometrically. The space P_0 has countable dimension and its completion is $\ell_1(\Gamma_0)$, so it is extremally projective by theorem 5.6. By propositions 2.1 and 2.8 the space $\ell_1^0(\Gamma) + V$ is extremally projective. □

By theorems 3.1 and 5.1, an extremally projective normed space is, up to an isometric isomorphism, a dense subspace of some $\ell_1(\Gamma)$ all of whose separable subspaces have countable algebraic dimension; by the theorem of Grønbæk it is also topologically isomorphic to $\ell_1^0(\Gamma)$. For countable Γ the first two conditions are also sufficient by theorem 5.8. We do not know whether this remains true for uncountable Γ.

Question 5.10. *Let Γ be an uncountable set and P be a dense subspace of $\ell_1(\Gamma)$ such that every separable subspace of P has countable algebraic dimension. Is P extremally projective? Is $d_{BM}(P, \ell_1^0(\Gamma)) = 1$?*

Let us comment on the possible approach to question 5.10. The proof of the theorem of Grønbæk consists of two steps: a transfinite construction which finds in every subspace Y of $\ell_1^0(\Lambda)$ a complemented subspace isomorphic to $\ell_1^0(\Lambda')$, where the cardinality of Λ' is the density character of Y [9, proposition 2.2], and the decomposition method of Pełczyński [9, proposition 2.3]. An inspection of the first step shows that it produces, for every $\varepsilon > 0$, a subspace which is $(1+\varepsilon)$-isomorphic to $\ell_1^0(\Lambda')$ and $(1+\varepsilon)$-complemented, so the constants are lost only in the second step. In the countable case theorem 5.5 replaces the decomposition method by a back-and-forth construction whose perturbations form a convergent infinite product. For uncountable Γ such a product can only converge if all but countably many of its factors act on pairwise disjoint blocks of coordinates, so an affirmative answer to question 5.10 would seem to require a decomposition of P into countably generated pieces supported on disjoint subsets of Γ, as in proposition 5.9.

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Norbert Nemesh, Faculty of Mechanics and Mathematics, Moscow State University, Moscow 119991 Russia

E-mail address: nemeshnorbert@yandex.ru